



TECHNICAL NOTICE

Third-party Runtime Components

MediaPipe provenance, modifications, checksums, licensing, and update procedure.

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MediaPipe 0.10.18 ARM64 wheel

The UNO Q runtime uses this tracked wheel:

```
python/worker-wheels/mediapipe-0.10.18-cp312-cp312-manylinux2014_aarch64.manylinux_2_17_aarch64.whl
```

Property	Value
Component	MediaPipe 0.10.18 for CPython 3.12, Linux ARM64
Upstream project	< https://github.com/google-ai-edge/mediapipe >
Upstream package	< https://pypi.org/project/mediapipe/0.10.18/ >
License	Apache License 2.0
PowerGlove repack SHA-256	f2617c0960eb35aaa58b76076a9ae629edbf7ec8901d5fab4b046c28d13fbe8d
Original PyPI wheel SHA-256	09cbf7dc1f9a2deeaac687e5f982836623def4cbd3e827d95f86f42450d2dd1

Modification notice

PowerGlove Vision repackaged the upstream wheel for its headless UNO Q worker. The Python package code and compiled MediaPipe binaries were not modified. The wheel dependency metadata was changed as follows, and its [RECORD](#) was rebuilt:

- removed the [jax](#) dependency declaration;
- removed the [jaxlib](#) dependency declaration;
- replaced [opencv-contrib-python](#) with [opencv-contrib-python-headless==4.10.0.84](#);
- omitted the upstream wheel's empty [mediapipe.libs](#) directory.

These changes avoid unnecessary JAX installation and GUI OpenCV dependencies on the UNO Q. The repacked wheel retains MediaPipe's Apache 2.0 license at [mediapipe-0.10.18.dist-info/LICENSE](#). Do not substitute the upstream wheel without retesting dependency resolution, camera startup, and hand tracking.

Google Hand Landmarker model

PowerGlove Vision uses Google's float16 Hand Landmarker task bundle.

Property	Value
Runtime path	models/hand_landmarker.task
Official download	< https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task >
SHA-256	fbc2a30080c3c557093b5ddfc334698132eb341044ccee322ccf8bcf3607cde1
Size	7,819,105 bytes

The model is not tracked in Git. Both `scripts/build-app-lab-package.sh` and `scripts/deploy-uno-q-wifi.sh` call `scripts/fetch-runtime-assets.sh`. That script downloads the model from Google's versioned URL and refuses to install it if the SHA-256 checksum differs. The generated App Lab ZIP contains the verified model so the installed application has a fixed runtime asset.

Google's MediaPipe repository is Apache-2.0 licensed and its official examples direct applications to this model URL. The model download does not place a separate model-specific license file in this repository. The PowerGlove Vision MIT License therefore should not be interpreted as licensing the Google model. Anyone publishing a prebuilt ZIP or otherwise redistributing the model should confirm that the intended distribution complies with Google's applicable model terms.

Updating either component

Treat a wheel or model update as a tested dependency change:

- 1 Record the official source URL, version, license, size, and SHA-256 here.
- 2 Update the pinned value in `scripts/fetch-runtime-assets.sh` when changing the model.
- 3 If repackaging another wheel, record every difference from upstream and retain its license files.
- 4 Build the App Lab ZIP and confirm it contains one wheel, one model, and only the root `sketch/` application sketch.
- 5 Test camera initialization, tracking, the Learn and Debug pages, and controller output on the UNO Q before publishing the package.